Part 23 — Single cells are not the average, and downstream is not 50 to 70 per cent

Two more criticisms. The first is right, and the connection the reviewer drew is sharper than the gap itself. The second I partly disagree with, and the disagreement is worth more to you in a defence than agreement would be — so chapter X2 interrogates the number rather than repeating it.

The audit

Topic	Mentions	Verdict
Stochastic expression and noise	64	Present, but almost all in chapter S6 as *bistability* — a circuit property, not a population one
Bimodal or all-or-none induction	14	**Thin**, and never linked to sorting
Single-cell heterogeneity	10	**Thin**
Persisters and non-growing subpopulations	**0**	**Absent**
Recovery train and unit operations	199	The module's most developed theme
Titer linked to separation cost	**332**	Strong — PP12 and EV7 reach the reviewer's own conclusion
Downstream share of installed capital	—	**Never computed** — the actual gap
Foaming and antifoam	11	**Thin** — mentioned, never quantified
Debris and filter blinding	21	**Thin** — PP10 computes release, not the filtration consequence

So on downstream the criticism is not that the module neglects it — it is the most heavily covered theme in the whole set — but that three specific things inside it were never computed. Two chapters, X1 and X2, and six problems, HN1 to HN3 and DS1 to DS3, taking the set to **155**.

Chapter X1 · extends chapters S6 and W3

Noise, bimodality and persisters: why a sorted hit may not reproduce

Every expression number in this module so far has been a population average. A sorter does not see averages — it sees single cells — and the gap between those two facts is a specific, computable way for a screening campaign to fail.

Where the noise comes from

Protein copy number in a single cell varies for two separable reasons. **Intrinsic noise** is the randomness of transcription and translation themselves; for a two-stage model with transcriptional bursts of size b,

$$CV^2 \approx 1/\langle P\rangle + [b/(1+b)]\cdot(1/\langle M\rangle)$$

Protein copies	mRNA copies	Burst size	CV	Share from bursting
30,000	26	12	**19%**	>99%
1,000	5	12	44%	99%
100	2	12	70%	98%

Two readings. At the 30,000 copies of part 14 chapter B8's strong gene, the Poisson term 1/⟨P⟩ is negligible and essentially **all the noise is transcriptional bursting** — so reducing noise means changing transcription dynamics, not raising the copy number. And a lowly expressed regulator at 100 copies carries a 70% coefficient of variation, which is why the repressor calculation in B1 came with a warning about stochastic effects.

Extrinsic noise — differences in cell size, ribosome content, plasmid copy number and cell-cycle position — is correlated across all genes in a cell and typically adds a CV of 0.2 to 0.4. Combining in quadrature, a 19% intrinsic noise plus 30% extrinsic gives a **total CV of 36%**, and the extrinsic part dominates. That matters for screening because extrinsic noise affects your reporter and your product *together*, so it is partly correctable by ratiometric measurement against a constitutive second reporter — the single most useful noise-reduction trick in a sorting experiment.

Bimodality: when the population splits

Noise makes a distribution wide. Positive feedback makes it **two distributions**. An inducible system whose inducer enters through a transporter that is itself induced has exactly that structure, and solving for steady states of $dx/dt = \alpha x^n/(K^n + x^n) + \text{leak} - x$ shows where it appears:

Hill coefficient	Steady states	Behaviour
1.0	one, at 0.539	Graded — every cell responds proportionally
1.5	three: 0.058, 0.077, 0.567	Marginally bistable
2.0	three: 0.022, 0.354, 0.644	**Bistable — the population splits**
3.0	three: 0.020, 0.464, 0.852	Strongly bistable

With two stable states and an unstable one between them, individual cells sit at one or the other and the population is a mixture. **The population mean is then a number no cell occupies.** With 20% of cells induced at a twentyfold higher signal, the mean reads 4.8 arbitrary units while every cell is at either 1 or 20.

The failure this causes in a sort

Here is the mechanism problem EV5 missed. EV5 attributed failed sorts to **sensor gaming** — a mutation that activates the reporter. But if the inducible system is bimodal, a top-1% gate selects **cells that happened to be in the induced state**, which is a phenotypic condition and not a genotype. Re-grow them and they re-split into the parent distribution: 20% induced before, 20% induced after. **The hit does not reproduce, and nothing is wrong with the strain.**

Even without bimodality, noise alone corrupts a gate. Simulating a library with a lognormal genotype effect and independent expression noise:

Genotype spread	Noise spread	Of a top-1% gate, truly top-1% genotype
0.6	0.2	66%

0.6	0.6	**27%**
0.3	0.6	**11%**
0.6	1.0	13%

When noise equals the genotype signal, **73% of a top-1% gate are lucky cells**, and when noise exceeds it, 89% are. The remedy is not a tighter gate — it is **sort, regrow, re-sort**, because regrowth re-randomises the noise while the genotype persists. Simulated, with noise equal to signal, the mean genotype of survivors rises from **1.87 standard deviations after one round to 3.03 after two and 3.79 after three**; with noise above signal it starts at 1.36 and reaches 4.09 by the third. Two rounds is usually the right number, and it costs one culture per round.

Persisters

A small fraction of any culture is in a non-growing, stress-tolerant state — typically 10^{-6} to 10^{-2} of cells depending on conditions. In a 100 m^3 fermenter at 10^{10} cells/mL that is **10^9 to 10^{13} cells**, and they have three properties that matter here: they **do not make product**, so they are a yield drag proportional to their fraction; they **survive stress**, including antibiotics and starvation; and they **regrow** when the stress lifts.

The consequence for selection is the important one. **Any selection that kills growing cells enriches persisters**, so a colony surviving an antibiotic challenge may be a persister rather than a mutant. The distinguishing test is heritability: re-grow the survivor without selection and re-challenge it. A mutant re-survives; a persister gives a culture as sensitive as the parent. That test belongs in every selection protocol and is routinely omitted, and it is the same logic as the sort-and-regrow control above — **a phenotype that does not breed true is not a genotype**.

Intrinsic noise	`CV² ≈ 1/⟨P⟩ + [b/(1+b)]/⟨M⟩; 19% at 30,000 copies, nearly all bursting`
Extrinsic	`adds 0.2–0.4 CV, correlated — correctable by a ratiometric second reporter`
Bimodality	`Hill ≥ 2 with leak gives two stable states; the mean is a value no cell holds`
Gate purity	`27% true genotype at equal noise and signal; 11% when noise dominates`
Remedy	`sort–regrow–re-sort: 1.87 → 3.03 → 3.79 standard deviations of genotype`
Persisters	`10⁻⁶–10⁻² of cells; any kill-based selection enriches them; test heritability`

What to open a book for. Elowitz et al. on distinguishing intrinsic from extrinsic noise (Science, 2002) — the two-colour experiment, read for the design; Ozbudak et al. on bimodality in the lac system; and a review of bacterial persistence. All three are short and the first two are the clearest experiments in the field.

Chapter X2 · extends chapter S1 and problem PP12

What downstream actually costs, and where "50 to 70 per cent" is true

The claim is widely repeated and product-class-specific. Computing it for two plants shows where it holds, and a candidate who can do that is in a better position than one who recites it.

The claim, tested on a commodity plant

Take PP12's propanediol: 30,000 t/yr at 135 g/L and 0.51 g/g. That is **222,000 m³ of broth a year**, which at 80% fill and a 60-hour cycle needs about **21 vessels of 100 m³**. Sizing the evaporation from duty rather than guessing: 6.4 kg of water per kg of product is 192,000 t/yr, which at 2.26 MJ/kg is 434,000 GJ/yr — an average duty of **15 MW**, and a multi-effect train for that is of order $40M.

Item	$M	Section
Fermenters, 21 × 100 m³	126.0	Upstream
Seed train	31.5	Upstream
Media preparation and sterilisation	4.0	Upstream
Utilities: air, steam, chilling	12.0	Upstream
Centrifuge and clarification	6.0	Downstream
Evaporation train, 15 MW	41.0	Downstream
Distillation and ion exchange	14.0	Downstream
Crystallisation and drying	8.0	Downstream
Tankage and offsites	10.0	Other
Total	**252.5**	**69% upstream, 27% downstream**

The same claim on a biologic plant

Now 100 kg/yr of a protein at 5 g/L — **20 m³ of broth a year**, which is two single-use 2 m³ vessels and nothing else upstream.

Item	$M	Section
Fermenters, 2 × 2 m³ single-use	3.0	Upstream
Seed train and media	2.0	Upstream
Utilities	2.0	Upstream
Harvest: centrifuge and depth filtration	2.5	Downstream
Capture chromatography	6.0	Downstream
Viral inactivation and two polish columns	9.0	Downstream
Ultrafiltration, diafiltration, fill-finish	7.0	Downstream
Facility: cleanrooms and quality control	14.0	Other
Total	**45.5**	**15% upstream, 54% downstream, 31% facility**

The verdict, and what to say in a defence

"50 to 70 per cent of capital is downstream" is right for biologics and wrong for commodities. The computed shares are **54% for the protein plant and 27% for the chemical plant**, and the reason is structural rather than accidental: a commodity needs enormous fermentation volume because the product is cheap and the titer is in grams per litre, while a biologic needs almost no volume and a long chromatographic train because purity is the specification. Repeating the number without the product class invites the question "for which product?" — and the answer is the interesting part.

Two refinements worth having ready. For a commodity, **separation dominates operating cost rather than capital**:

at 135 g/L the steam for water removal is $0.12/kg against $0.78/kg of sugar, and at 50 g/L it is $0.34 — so the titer argument from PP12 and EV7 is an *operating* cost argument, which is where it belongs. And the biologic's capital is dominated as much by the **facility** — cleanrooms, air handling, quality control — at 31% as by any single unit operation, which is why single-use equipment and facility flexibility matter more than column efficiency in that business.

Two unit-operation failures the module had only mentioned

Foaming. Protein-rich broth foams, and the standard response, chemical antifoam, promotes bubble coalescence — which is exactly what reduces interfacial area. Taking an illustrative relation of k_La falling as $1/(1 + 0.004 \cdot ppm)$:

Antifoam	Relative k_La
10 ppm	0.96
50 ppm	0.83
200 ppm	**0.56**

So heavy antifoam dosing costs **44% of oxygen transfer** — it solves a foam problem by creating the oxygen problem of chapter T4, and the interaction is why antifoam is dosed on demand from a foam probe rather than added to the batch. The alternative, a mechanical foam breaker, costs power and nothing else. The failure mode if neither works is a **foam-out**: foam carries cells into the exhaust filter, blinds it, and raises headspace pressure — in a 100 m³ vessel, 15% foam is **15 m³ to disengage**, which the headspace does not have.

Debris. PP10 computed product release against homogeniser passes and stopped there. Each pass also **fragments the debris further**, and sub-micron debris blinds a depth filter by cake resistance. With filtrate flux falling as $1/(1 + 0.8 \cdot n)$:

Passes	Relative flux	Filter area needed
1	0.56	1.8×
3	0.29	**3.4×**
5	0.20	5.0×

So PP10's optimum of three passes, chosen on product quality alone, carries a **3.4-fold filter area** penalty that the product-release calculation never saw. The full optimum is lower than either calculation alone suggests, and the clean way to state it is that **disruption and clarification must be optimised together** — which is the general form of this chapter's argument and of chapter S1's.

Commodity capital	`69% upstream, 27% downstream at 30,000 t/yr and 135 g/L`
Biologic capital	`15% upstream, 54% downstream, 31% facility at 100 kg/yr and 5 g/L`
The claim	`"50–70% downstream" is a biologics number; for a commodity it is operating cost, not capital`
Antifoam	`200 ppm costs 44% of kLa; dose on demand or use a mechanical breaker`
Debris	`3 homogeniser passes need 3.4× the filter area of one`

The capital figures above are order-of-magnitude estimates built from scaling rules and my own judgement, not from vendor quotations — they are good enough to establish which section dominates and not good enough to put in a proposal. A real estimate comes from the NREL-style design reports referenced in the reading list. The antifoam and debris relations are illustrative forms chosen to show the shape; the coefficients are yours to measure.

Problems HN1–HN3 and DS1–DS3

HN1 · chapter X1 · 1.5 h

Where the noise in your reporter comes from

1. Using $CV^2 = 1/\langle P\rangle + [b/(1+b)]/\langle M\rangle$, compute the CV at 30,000, 1,000 and 100 protein copies with the mRNA and burst values given.
2. What share of the variance is Poisson and what share is bursting in each case?
3. Add extrinsic noise of 0.2 and 0.3 in quadrature to the 30,000-copy case.
4. Which component can a ratiometric measurement remove, and why?

Worked solution. CVs are **19%, 44% and 70%**, and bursting accounts for over 98% of the variance in all three — the Poisson term $1/\langle P\rangle$ matters only below a few hundred copies. Adding extrinsic noise gives **28% at 0.2 and 36% at 0.3**, so above about 10,000 copies the extrinsic component dominates the total.

A ratiometric measurement — your reporter divided by a constitutive second fluorophore in the same cell — removes the **extrinsic** component, because cell size, ribosome content and plasmid copy number affect both reporters together and cancel in the ratio. It cannot remove intrinsic noise, which is independent for each gene by construction. That is exactly what the two-colour experiment of Elowitz and colleagues was designed to separate, and it is why a sorting experiment with a single colour is measuring cell size as much as pathway activity.

What it teaches: noise has two components with different remedies, and the one a sort can correct for is the one that dominates at useful expression levels.

HN2 · chapter X1 · 2 h

The hit that will not reproduce

1. Solve for the steady states of $dx/dt = \alpha x^n/(K^n + x^n) + \text{leak} - x$ at Hill coefficients of 1, 1.5, 2 and 3, with $\alpha = 1$, $K = 0.5$, leak = 0.02.
2. With 20% of cells induced at a twentyfold signal, compute the population mean and say where the cells actually are.
3. What is a top-1% gate selecting, and what happens when you re-grow it?
4. Design the control that distinguishes this from a real genotype, and distinguish it from EV5's sensor gaming.

Worked solution. One steady state at n = 1 (0.539), three from n = 1.5 upward — at n = 2 they are 0.022, 0.354 and 0.644, with the middle one unstable, so cells occupy the low or high state and the population is a mixture. At 20% induced and twentyfold signal the mean is **4.8**, a value **no cell holds**.

A top-1% gate is therefore selecting **cells in the induced state**, not cells with a better genotype. Re-grown, they re-split to 20% induced and the apparent improvement vanishes. The control is **sort, regrow a clonal culture, and re-measure the distribution**: if it reproduces the parent's bimodal shape, you selected a state; if the whole distribution has shifted, you selected a genotype. That differs from EV5's sensor gaming in both mechanism and test — gaming is a heritable mutation in the reporter, so it *does* breed true and is caught by assaying **product** rather than fluorescence, whereas a phenotypic state does not breed true at all and is caught by re-measuring the distribution.

What it teaches: a sort can select a phenotypic state rather than a genotype, and the diagnostic is whether the distribution reproduces, not whether the clone is bright.

HN3 · chapter X1 · 1.5 h

How much of your gate is luck, and how many rounds fix it

1. For genotype and noise spreads of (0.6, 0.2), (0.6, 0.6), (0.3, 0.6) and (0.6, 1.0), compute what fraction of a top-1% gate truly has a top-1% genotype.
2. Why does a tighter gate not help?
3. Simulate sort, regrow and re-sort for three rounds at equal noise and signal, and report the mean genotype of survivors.
4. How many rounds would you run, and what does each cost?

Worked solution. True-genotype fractions are **66%, 27%, 11% and 13%** — so when noise equals signal, nearly three-quarters of your gate is lucky cells, and when the genotype spread is small relative to noise it is nine-tenths. A tighter gate does not help because it selects more extreme *observed* values, and the most extreme observations are the ones

with the largest noise contribution: narrowing from the top 1% to the top 0.1% makes the precision worse, not better.

Sort–regrow–re-sort works because regrowth **re-randomises the noise while the genotype persists**. Simulated at equal noise and signal, the mean genotype of survivors is **1.87 standard deviations after one round, 3.03 after two and 3.79 after three**; with noise above signal, 1.36, 2.48 and 4.09. I would run **two rounds** — the second adds more than the third and each costs one overnight culture plus one sorter session, which against the alternative of characterising hundreds of false positives is trivially worthwhile.

What it teaches: gate precision is set by the ratio of genotype spread to noise, and the fix is more rounds rather than a narrower gate.

`DS1` · chapter X2 · 2 h

Compute the capital split for your own process

1. For 30,000 t/yr at 135 g/L, compute annual broth volume, vessels required at 80% fill and a 60-hour cycle, and the evaporation duty from the water load.
2. Build an item list and compute the upstream and downstream shares of capital.
3. Repeat for 100 kg/yr of a protein at 5 g/L.
4. Is "50 to 70 per cent of capital is downstream" true? State the conditions under which it is.

Worked solution. The commodity needs **222,000 m³ of broth a year**, about **21 vessels of 100 m³**, and an evaporation duty of **15 MW** from 192,000 t of water. Its capital comes out **69% upstream and 27% downstream** on a $252M total. The biologic needs **20 m³ a year** — two small single-use vessels — and comes out **15% upstream, 54% downstream and 31% facility** on $45M.

So the claim is **true for biologics and false for commodities**, and the reason is structural: a cheap product at grams per litre needs enormous fermentation volume, while an expensive product at grams per litre needs almost none and a long purification train instead. The conditions under which it holds: **high product value, low volume, purity-driven specification, and a chromatographic train**. For a commodity the honest statement is different — separation dominates *operating* cost, at $0.12/kg of steam against $0.78/kg of sugar at 135 g/L, rising to $0.34 at 50 g/L — which is where PP12's and EV7's titer argument actually lands.

What it teaches: a widely quoted cost share is product-class-specific, and computing it for two classes is a better answer than quoting it for one.

`DS2` · chapter X2 · 1 h

Foam, and the oxygen it costs you

1. With k_La falling as $1/(1 + 0.004 \cdot ppm)$, compute the relative transfer at 10, 50 and 200 ppm of antifoam.
2. Why does antifoam reduce k_La at all?
3. What volume of foam is 15% of a 100 m³ vessel, and what does a foam-out do?
4. Compare chemical and mechanical foam control, and say how you would dose.

Worked solution. Relative k_La is **0.96, 0.83 and 0.56** — so heavy dosing costs **44% of oxygen transfer**, which for an oxygen-limited process is worse than the foam. Antifoam works by **promoting bubble coalescence**, and coalescence is precisely what destroys the interfacial area that k_La depends on: the mechanism that breaks the foam is the mechanism that cuts transfer. They are not separable.

Fifteen per cent of a 100 m³ vessel is **15 m³ of foam**, which the headspace does not have — the consequence is a **foam-out**: foam carries cells into the exhaust filter, blinds it, raises headspace pressure, and in the worst case forces sterile boundary failure. Chemical control is cheap, immediate and costs transfer; mechanical control — a foam-breaking impeller or a centrifugal breaker — costs power and capital and no transfer. The practice that follows: **dose on demand from a conductivity foam probe rather than batching antifoam in**, since batched antifoam pays the k_La penalty for the whole run to solve a problem that occurs for part of it.

What it teaches: foam control and oxygen transfer are the same physics with opposite signs, so the control strategy matters more than the agent.

`DS3` · chapter X2 · 1.5 h

Debris: the cost PP10 did not count

1. With filtrate flux falling as $1/(1 + 0.8 \cdot n)$ with homogeniser passes n, compute relative flux and filter area at 1, 3 and 5

passes.

2. PP10 chose three passes on product quality. What does that choice cost in clarification?
3. Combine the two and say qualitatively where the joint optimum lies.
4. What alternatives to homogenisation change this trade, and what do they cost?

Worked solution. Relative flux is **0.56, 0.29 and 0.20**, so area required is **1.8×, 3.4× and 5.0×** a baseline. PP10's three-pass optimum therefore carries a **3.4-fold filter area** penalty, or equivalently 3.4 times the filtration time, which its product-release calculation never saw.

Combining them: product recovery rises steeply from one to two passes and flattens, while filtration cost rises roughly linearly, so the **joint optimum is lower than PP10's three passes** — probably two — and the right way to find it is to cost both curves in the same units rather than optimise either alone. That is the general lesson: adjacent unit operations are coupled, and an optimum computed for one is not an optimum. Alternatives that change the trade: **bead milling**, which is worse for debris; **enzymatic or chemical lysis**, which is gentler and leaves larger fragments that filter better, at the cost of the lysing agent as an impurity; and **flocculation before filtration**, which aggregates the fines and recovers much of the flux for the cost of a polymer that must then be cleared.

What it teaches: optimising a unit operation in isolation exports its cost to the next one, and disruption exports to clarification.

Closing notes to part 23

On the two criticisms

The first was right and I had missed the connection. Noise was in the module as bistability — a circuit property — and not as population heterogeneity, and the link to sorting was absent. That link is the valuable part: a bimodal inducible system makes a top-1% gate select a *state* rather than a genotype, the hit does not reproduce, and the control is different from the one EV5 prescribes for sensor gaming. Add that noise alone leaves only **27% of a top-1% gate** truly top-1% when it equals the signal, and the practical conclusion is that **two rounds of sort-and-regrow should be the default**, not an afterthought.

The second was partly right about a theme that was already the most developed in the module. Downstream appears 199 times and the titer-to-separation link 332. What was genuinely absent was the capital split itself, and computing it gives a result that contradicts the suggestion as stated: **27% downstream for a commodity against 54% for a biologic**. I would rather you went into a defence able to compute that than able to quote "50 to 70 per cent", because the follow-up question — "for which product class?" — is the one a committee asks.

The two unit-operation gaps inside that theme were real and now computed: antifoam at 200 ppm costs **44% of oxygen transfer**, and PP10's three-pass optimum costs **3.4 times the filter area** it never counted.

The numbers worth keeping

- At 30,000 protein copies the CV is **19%**, and over 98% of the variance is transcriptional bursting.
- Extrinsic noise of 0.3 takes the total CV to **36%** and is removable by a ratiometric reporter.
- A Hill coefficient of **2** with a small leak splits the population into two states.
- At equal noise and signal, **27%** of a top-1% gate truly has a top-1% genotype; at noise above signal, **13%**.
- Sort–regrow–re-sort: **1.87 → 3.03 → 3.79** standard deviations of genotype over three rounds.
- Persisters are 10^{-6} to 10^{-2} of cells; any kill-based selection enriches them, so test heritability.
- Downstream capital share: **27% commodity, 54% biologic**; the biologic's facility alone is 31%.
- Antifoam at 200 ppm: **0.56** relative k_La.
- Three homogeniser passes: **3.4×** the filter area of one.

On the assumptions

The simulations and balances are computed. The inputs to challenge: the burst size of 12 and the extrinsic CV range, both recalled; the capital figures, which are order-of-magnitude estimates from scaling rules and judgement rather than quotations, and which are adequate to establish which section dominates and not to put in a proposal; and the antifoam and debris relations, whose *forms* are standard and whose coefficients are illustrative. For real capital numbers, the NREL-style design reports in the reading list are the source.

Where the set stands

Source	Problems
Curriculum appendix A1–W3	54
Parts 15–22	93
Part 23 HN1–HN3, DS1–DS3	6
Total in the tracker	**155**